

Roshan Shukla

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PROFESSIONAL SUMMARY

Computer Science graduate with production experience building distributed backend systems, concurrent software, and scalable APIs. Experienced in Java, Spring Boot, Redis, PostgreSQL, and production performance optimization through concurrency, caching, and distributed systems. LeetCode Knight (1890).

EDUCATION

B.Tech – Computer Science & Engineering (AI/ML) Nov 2022 – Jun 2026
Dr. A.P.J. Abdul Kalam Technical University (AKTU), Uttar Pradesh **CGPA: 8.09 / 10**

EXPERIENCE

Founding Software Engineer *SymptomWise Pvt. Ltd* Oct 2025 – Apr 2026

- Built a concurrent bounded priority scheduler with dynamic aging to eliminate request starvation under burst traffic.
- Engineered a two-level Redis cache using exact-match and semantic caching, reducing production p99 latency by **82%**, from 1.2s to 213ms, and achieving a **61% cache hit rate**.
- Enforced thread-safe tenant isolation to guarantee per-hospital data integrity across concurrent request execution.
- Validated production readiness via Locust with **5,800+** requests and 100 concurrent users, achieving zero request failures.

Software Engineering Intern *GanakGyan Technologies* Jun 2025 – Sep 2025

- Architected backend systems for a geospatial platform managing **50K+** location records, built GraphQL APIs and designed the MongoDB schema.
- Implemented RBAC and JWT authentication across REST and GraphQL services, enforcing scoped access with Swagger documentation.
- Migrated the build system from Maven to Gradle, standardizing dependency management and development workflows.

Software Engineering Intern *Karyanest LLP* Feb 2025 – Apr 2025

- Built a Python ingestion pipeline to detect and reject malformed and duplicate pincode records before MySQL load.
- Implemented RBAC and JWT-based authentication with Spring Security, enforcing role-scoped access across all REST endpoints.
- Improved regional address lookup performance via targeted MySQL index optimization across the address dataset.

PROJECTS

IntervYou : Real-Time Distributed Interview Platform Live | GitHub
Next.js • TypeScript • Yjs • WebRTC • Redis • PostgreSQL

- Built a distributed real-time interview platform with custom WebRTC signaling, P2P media, Yjs collaboration, Redis, and PostgreSQL, achieving **47ms p50** and **94ms p99** realtime synchronization latency with **100% delivery success**.
- Designed an atomic matchmaking lock using Redis **SET NX**, preventing duplicate lobby claims and rejecting late-binding clients with HTTP 409.
- Implemented field-level optimistic concurrency control to resolve concurrent session updates without row-level database locking.
- Built Redis write-behind session persistence with PostgreSQL snapshots and a **30-second** reconnect grace period for reliable session recovery after transient disconnects.
- Engineered a fault-tolerant code execution pipeline using Piston with Judge0 fallback and guarded execution timeouts for untrusted code execution.

NarrateLoop : Autonomous Reddit-to-Video GenAI Pipeline Live | Docs | GitHub
Python • FastAPI • FFmpeg (NVENC) • PyTorch • Whisper • Linux (systemd) • Oracle Cloud (OCI)

- Architected a 24/7 autonomous GenAI pipeline on Oracle Cloud integrating Suno Bark, Whisper alignment, and Gemini NLP with restartable asset tracking.
- Implemented GPU memory monitoring via GPUtil with automated CPU fallback on CUDA OOM for unattended batch processing.
- Rendered GPU-accelerated video via FFmpeg NVENC encoding with ASS word-synchronized subtitle overlays.
- Developed a FastAPI telemetry service with interactive Swagger /docs, Nginx SSL proxy, and Telegram bot supervision.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, JavaScript
Web: HTML, CSS, Next.js, REST APIs, GraphQL, WebSockets
Backend: Spring Boot, REST APIs, GraphQL, WebSockets, Redis, Concurrent Programming, Distributed Systems
Databases: PostgreSQL, MongoDB, MySQL, Prisma ORM
Core CS: Data Structures & Algorithms, OOP, Operating Systems, Computer Networks, DBMS
Tools: Docker, Linux, Git, GitHub Actions, Maven, Gradle

ACHIEVEMENTS

- Solved **700+** DSA problems (**550+** LeetCode and **180+** GeeksforGeeks) **LeetCode Knight (1890)**.
- IEEE Student Branch Secretary** led a 15-member team across technical events and initiatives.
- 2nd Runner-Up at TechYuva 10.0, competitive engineering cohort.
- Certifications:** AWS ML Foundations, Microsoft Generative AI, Machine Learning (IIT BHU).